

How to Publish Your Research Data in 5 Easy Steps

Open Science Team | Bern University Library
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DOI: [10.5281/zenodo.10657089](https://doi.org/10.5281/zenodo.10657089)



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Publish your data in 5 easy steps

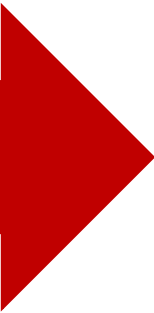
1. Select
data

2. Choose
repository

3. Prepare
your data

4. Deposit

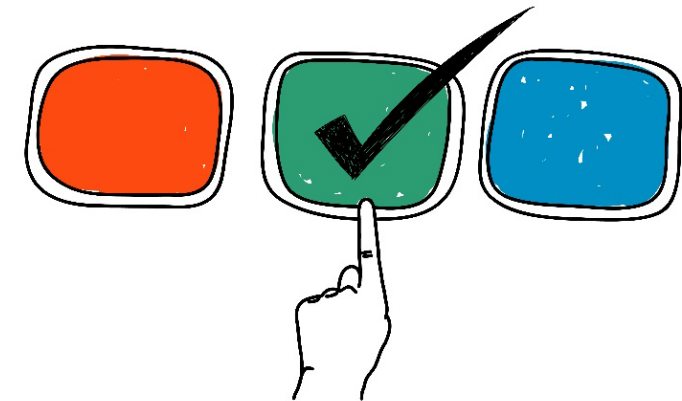
5. Link
your
datasets



u^b 1. Select data

Questions to consider

- Are there any policies you have to comply with?
- Are there any legal regulations?
- Are data subject to contractual agreements?
- Are data necessary for validation of published results?
- Does your data have reuse potential?
- Which of your data are reusable?
- How difficult is it to replicate your data?
- Can data be further exploited?



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1. Select data

General Do's and Dont's

Share

- all data that are necessary to reproduce / replicate / understand your published research findings
- even more data (e.g. raw data if possible), code and supplementary material to optimize re-use potential
- documentation explaining provenance, context and structure of your data

Don't share

- unanonymized data containing personal information
- data without informed consent of study participants
- data containing copyright-protected or classified information without written permission



Check options to restrict access to data on repository!



If in doubt: Check with the legal support at your institution!

u^b 2. Choose a repository

What are repositories?

Research data repositories are the place where you make your data accessible online.

The image shows a screenshot of the Zenodo website, which is a research data repository. The top navigation bar is blue with the 'zenodo' logo. Below it, the 'Recent uploads' section displays a list of datasets. The first dataset is 'Inanspruchnahme von Impfsurveillance' by Rieck, Thorsten and Feig, M., uploaded on December 11, 2023. The second dataset is 'Organic soil surveys' by De Dobbelaer, Tom and Herremans, J., also uploaded on December 11, 2023. The third dataset is 'LISA Constants' by Bayle, Jean-Baptiste and Le, uploaded on December 11, 2023. To the right of the Zenodo interface, there is a separate window showing the 'BORIS Portal' (Bern Open Repository and Information System) interface. This portal has a red navigation bar with links for Home, Projects, Funding, Research Data, Organizations, and About. It features a search bar with the placeholder 'Search term...' and a red 'Search' button. Below the search bar, the 'Last added' section lists several datasets, including 'TRP, WMupdating, and g' by Troche, Stefan; Makowski, Lisa Michaela; Pahud, Olivier; Rammsayer, Thomas; 'Dataset for large regional variation in cardiac closure procedures to prevent ischemic stroke in Switzerland. A population-based small area analysis.' by Stoller, Nina Danielle; Wertli, Maria Monika; Haynes, Alan; Chiolero, Arnaud; Rodondi, Nicolas; Panczak, Radoslaw; Aujesky, Drahomir; 'Intestinal Dysbiosis as an intraoperative predictor of Septic Complications: Evidence from Human Surgical Cohorts and Preclinical Models of Peritoneal Sepsis' by Beldi, Guido; Candinas, Daniel; Salm, Lilian; Yilmaz, Bahtiyar; Zwicky, Simone Nora; Spari, Daniel; and 'Daily temperature and sea-level pressure fields for the Southern Hemisphere 1901-1930' by Brönnimann, Stefan.

u^b 2. Choose a repository

Subject-specific data repository

Advantages

- Visibility within community
- Offer subject-specific metadata schemas that make finding and understanding data easier
- Meet technical requirements for specific data types

Examples

- [GenBank](#) (Genome data)
- [Pangaea](#) & [NOAA](#) (Earth & Environmental Science)
- [SWISSUbase](#) (Social Sciences, Linguistics, for CH)
- [DaSCH](#) (Humanities, for CH)

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2. Choose a repository

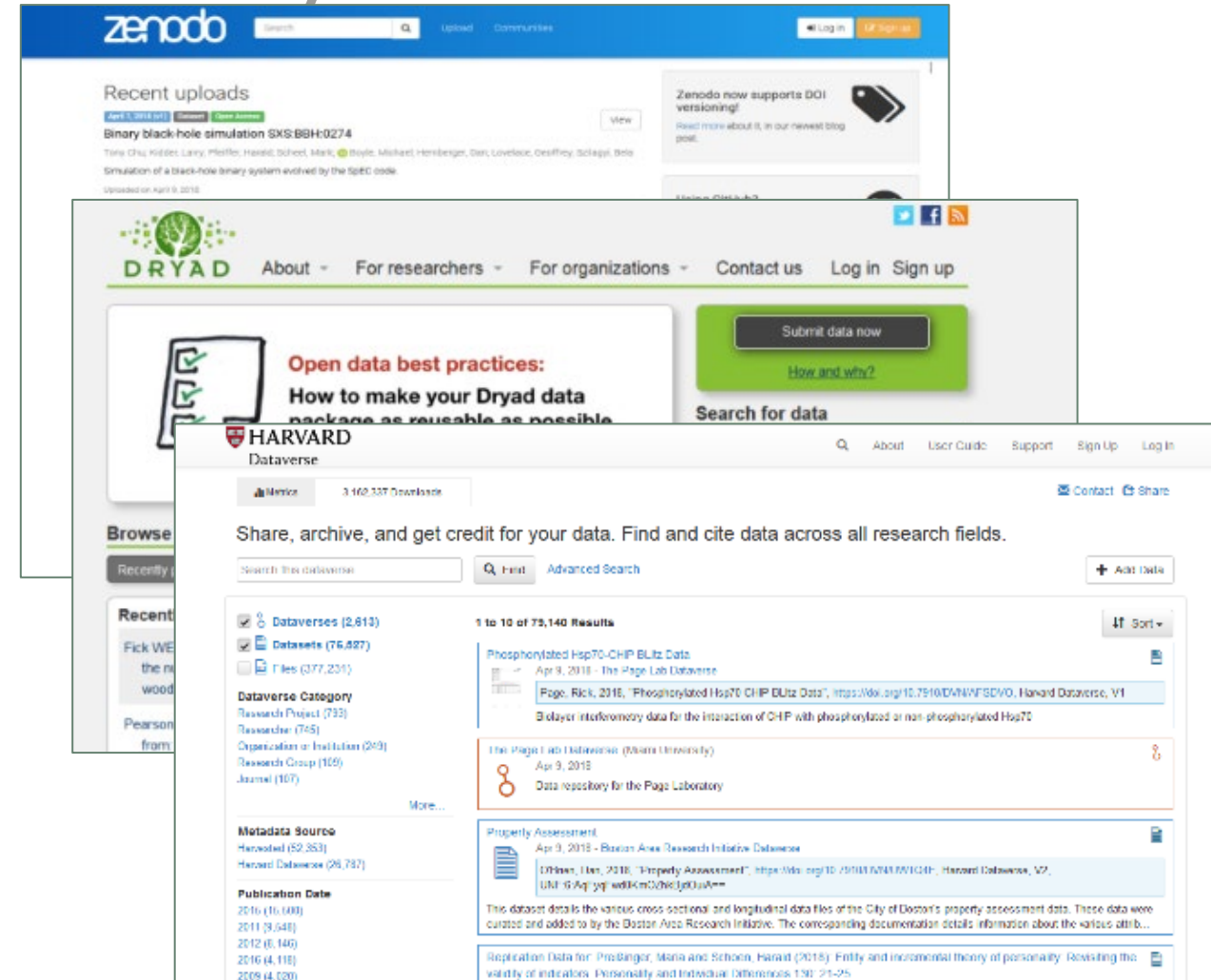
Generalist data repository

Advantages

- Open to all disciplines
- Generally easy to use

Examples

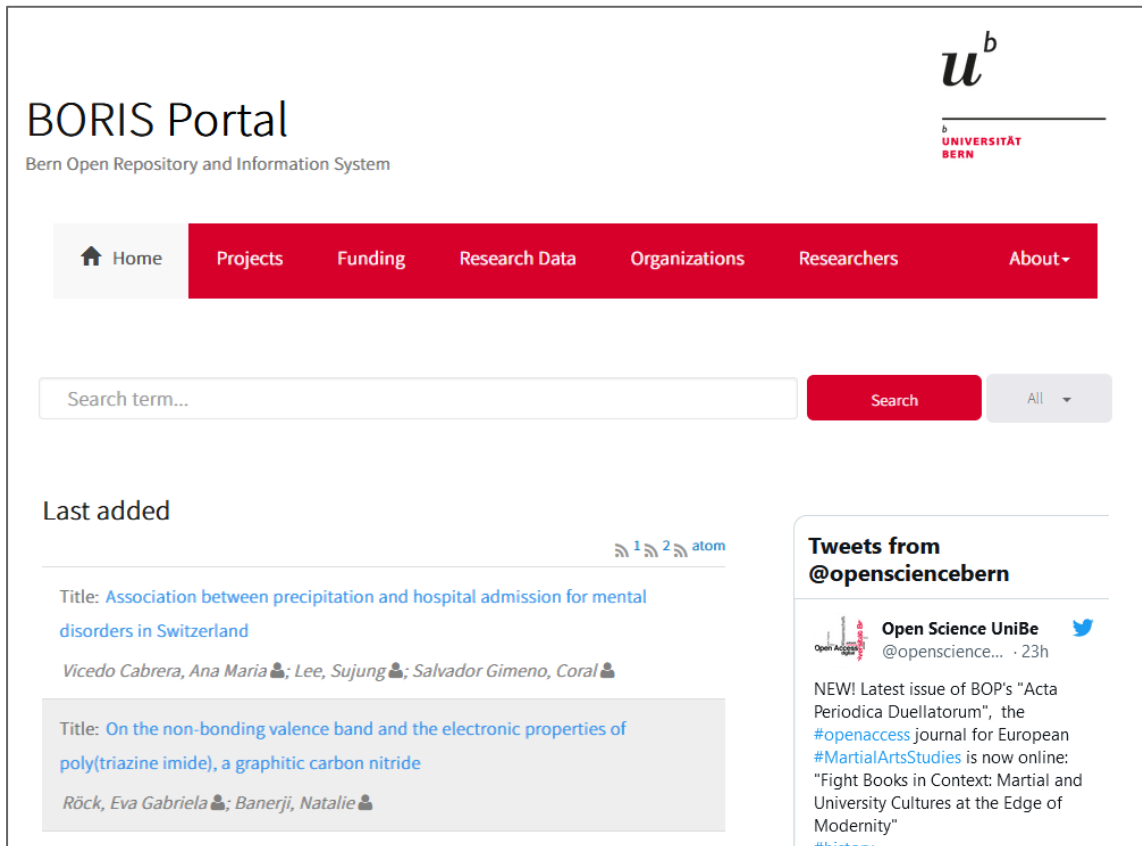
- [Zenodo](https://zenodo.org/)
- [Harvard Dataverse](https://dataverse.harvard.edu/)
- [Dryad](https://www.dryad.org/)



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2. Choose a repository

Institutional data repository



At the University of Bern:

BORIS Portal: Research data and project information

<https://boris-portal.unibe>

Advantages

- All data in the same place
- In-house service: deposit check, support



Ask your research data support for your institutional options!

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2. Choose a repository

Choose a suitable repository

1. Use discipline-specific repositories

- Use one you or your colleagues know/use
- Use the resources for finding a repository
- Ask your research data support team

2. If there are none, use a generalist repository (e.g., Zenodo)

3. Make sure the repository you choose complies with requirements and/or is listed by funder/publisher

Resources for finding a repository

Repository registries

- Re3data: www.re3data.org
- FairSharing: <https://fairsharing.org/>

SNSF checklist ([Link](#))

Lists of accepted repositories

- [SNSF repository list](#) (non-exhaustive)
- [Open Research Europe list](#)
- [Nature](#), [Plos One](#)

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2. Choose a repository

Code repositories

- Use code repositories like GitHub, GitLab, or Bitbucket for development
- Release a stable version to a repository to get a DOI and enable FAIR data
- Use the [Zenodo-Github-Integration](#) to do this faster
- Use a license selector (e.g., [this one](#) or [this one](#)) to find the right Open-Source license for your code

zenodo Search records... Q Communities My dashboard Log in Sign up

Published February 12, 2024 | Version 0.14.9 Software Open

ComPWA/ampform: Automatically generate symbolic amplitude models for Partial Wave Analysis

Fritsch, Miriam¹; Pflüger, Stefan¹; de Boer, Remco E.¹; Jäger, Sebastian¹; Wollenberg, Leonard¹; Gradl, Wolfgang²; Peters, Klaus³ Show affiliations

AmpForm is a Python library of spin formalisms and dynamics with which you can automatically formulate symbolic amplitude models for Partial Wave Analysis. The resulting amplitude models are formulated with [SymPy](https://www.sympy.org/en/index.html) (a Computer Algebra System). This not only makes it easy to inspect and visualize the resulting amplitude models, but also means the amplitude models can be used as templates for faster computational back-ends (see [TensorWaves](https://github.com/ComPWA/tensorwaves)).

Notes

If you use this software, please cite it as below.

Files

ComPWA/ampform-0.14.9.zip

ComPWA/ampform-0.14.9.zip

ComPWA-ampform-df2dd8e

- constraints
- gitignore 11 Bytes
- py3.10.txt 4.3 kB
- py3.11.txt 4.2 kB
- py3.12.txt 4.2 kB
- py3.7.txt 4.5 kB
- py3.8.txt 4.4 kB

Versions

Version	Date
Version 0.14.9 10.5281/zenodo.10650291	Feb 12, 2024
Version 0.14.8 10.5281/zenodo.10426368	Dec 22, 2023
Version 0.14.7 10.5281/zenodo.5566862	Dec 15, 2023
Version 0.14.6 10.5281/zenodo.7933052	May 13, 2023
Version 0.14.5 10.5281/zenodo.7402081	Dec 5, 2022

View all 9 versions

Cite all versions? You can cite all versions by using the DOI 10.5281/zenodo.5526648. This DOI represents all versions, and will always resolve to the latest one. Read more.

External resources

Available in

ComPWA/ampform
Release: 0.14.9

Indexed in

doi.org/10.5281/zenodo.5526648

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Share Your Data in 5 Steps

1. Select
data

2. Choose
repository

3. Prepare
your data

4. Deposit

5. Link
your
datasets



3. Prepare your data

Check for repository guidelines

DEPOSIT YOUR SOCIAL SCIENCES DATA ON SWISSUBASE: BASIC REQUIREMENTS

FORS uses the SWISSUbase online platform to facilitate research data management and data archiving for all social science disciplines in a secure and controlled environment.

It is a non-commercial, national, public and transdisciplinary archiving service solution that provides controlled access to research data, ranging from open access to restricted access, in a secure and controlled environment.

In order to deposit social science data in [SWISSUbase](https://www.swissubase.ch) you must meet the following requirements.

Have a SWITCH edu-ID account requirement to use swissubase

- Log in to SWISSUbase using your edu-id account: <https://www.swissubase.ch>
- If you do not have a SWITCH edu-ID account, you can get one here: <https://www.switch.ch/edu-id/>

Requirements concerning the data

The deposited data can be quantitative and qualitative and must meet the following requirements:

- Represent works of general interest for social science research with a Swiss component
- Have been collected in accordance with prevailing data protection laws and permit reuse by way of sufficient anonymisation and/or explicit consent for sharing
- Have been sufficiently documented so that the context of the data can be understood and the data can be reused

These points imply for the depositor to prepare the data for sharing and thus to consider the following aspects of data preparation.

Requirements concerning data preparation

- Check whether **copyright** permission should be sought with regard to data **ownership**
- Provide an appropriate level of **anonymisation** so that the participants can not be recognized
- Consistently use meaningful and self-explanatory **file names, variable names**, codes and abbreviations
- Make sure there is consistent handling and labeling of **missing values**
- Remove all your own **temporary or administrative variables** created for internal purposes
- Check our [transcription format](#) for qualitative textual data
- Use our preferred **file formats**: .sav; .por; .xls; .xlsx; .csv; pdf; .rtf; .txt; doc; .docx; .odt; .jpeg; .jpg; .jp2; .tif; .tiff; .raw; .bmp; .png; .mp3; .flac; .wav; .aac; .mp4; .mov; .avi; .mj2
- Check the **size** of your data file(s) and contact our data service if your data files are larger than 1 TB

Consult our guides and documentation

- [Sharing data: a step-by-step checklist](#)
- [Preparing your data for deposit in SWISSUbase: in brief](#)
- [Preparing your data for deposit in SWISSUbase: in detail](#)
- [SWISSUbase user guide](#)
- [SWISSUbase advantages](#)

u^b 3. Prepare your data

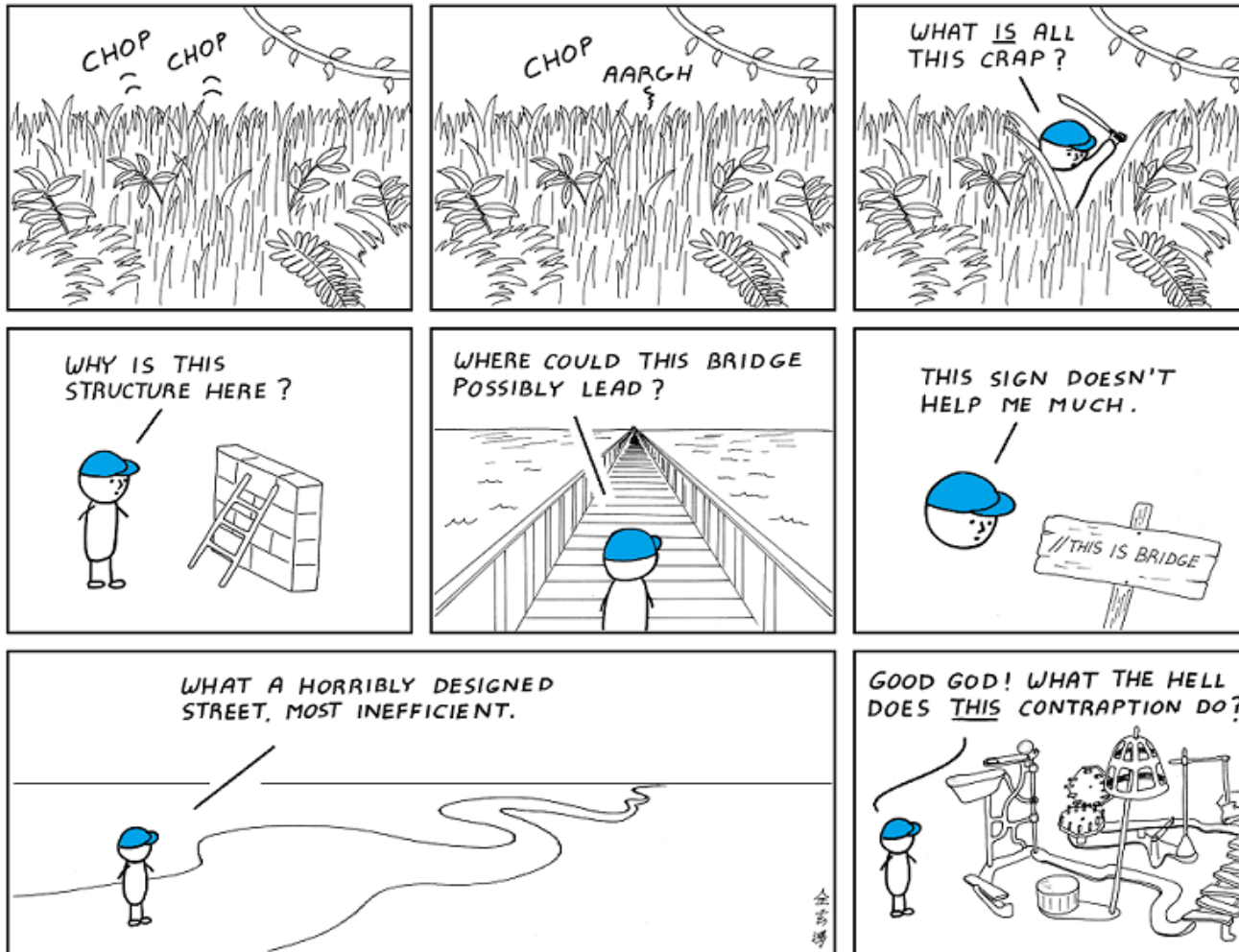
General considerations

- **Sensitive data:** make sure your data is fully anonymized/pseudonymized
- **Documentation:** make sure you add [sufficient documentation](#) codebooks, questionnaires, [ReadMe Files](#), etc)
- **File formats:** use open file formats as far as possible e.g. csv, txt, PDF/A (see [ETH](#))
- **File names:** check if your files have [meaningful names](#)
- **File size:** check limitation of repository
- **Table data:** variables / column titles and abbreviations are explained
- **Code and scripts:** Don't forget to also make code and scripts available

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3. Prepare your data

A word on documentation



«O.P.C» by [Abstruse Goose](#) is licensed under [CC BY-NC 3.0 US](#)

I hate reading
other people's code.

Try to document all information necessary to reproduce your results.

Including:

- Software version used and details of accessibility
- Any variable parameters that could impact the outcome of the results

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4. Deposit

Access options - Zenodo

Different options for different needs:

Open – Standard: for data that contain no personal or disclosive information.

Embargoed – if data should not be available before a certain date.

Restricted – e.g. for data that contain no personal information, however, is considered to contain a residual risk of disclosure.

Example: Zenodo

Visibility *

Files only

Public

Restricted



Public
The record and files are publicly accessible.

Options

☐ **Apply an embargo** ⓘ
Record or files protection must be restricted to apply an embargo.

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4. Deposit

Published is published

- **Double check** - make sure you have uploaded the correct files
- **Describe** your data as **detailed** as possible and use standards
 - Title: descriptive, important search words at the beginning, no special characters if possible
 - Abstract: important information at the beginning, repeat keywords and use synonyms
 - Keywords: use controlled vocabularies
- Make sure to **give credit** to all contributors
- **Link** to publication and other data sources (update entry if publication is not yet published)
- **Choose license**

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4. Deposit

Licenses - What are they good for?

If you create output:

- **Clear statement** that your work can be reused by others
- **Conditions** that apply if someone would like to reuse your work
- **Protecting** your rights
- **Comply** with funding requirements

If you would like to reuse output:

- Easy way to see if **reuse is possible**
- **Conditions** for reusing a work are clearly defined

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4. Deposit

One option – Creative Commons

CREATIVE COMMONS LICENSES		 COPY & PUBLISH	 ATTRIBUTION REQUIRED	 COMMERCIAL USE	 MODIFY & ADAPT	 CHANGE LICENSE
	PUBLIC DOMAIN	✓	✗	✓	✓	✓
	CC BY	✓	✓	✓	✓	✓
	CC BY-SA	✓	✓	✓	✓	✗
	CC BY-ND	✓	✓	✓	✗	✗
	CC BY-NC	✓	✓	✗	✓	✓
	CC BY-NC-SA	✓	✓	✗	✓	✗
	CC BY-NC-ND	✓	✓	✗	✗	✗

 You can redistribute (copy, publish, display, communicate, etc.)

 You have to attribute the original work

 You can use the work commercially

 You can modify and adapt the original work

 You can choose license type for your adaptations of the work.

- 4 building blocks – 6 Licenses
- Recommendation CC0 or CC BY
- Use latest Version (4.0) for data
- CC-License chooser:
<https://chooser-beta.creativecommons.org/>
- Give attribution if you reuse data
- Do **not** use for restricted data or code

u^b 5. Link your datasets

- Add a data availability statement to your article including a DOI (some repositories allow you to reserve a DOI)
- Examples from [Cambridge University Press](#) and [Springer Nature](#)

- Link your articles, datasets, code

Related identifiers:

Cited by

[arXiv:2004.04813](#) (Preprint)

Compiled by

<https://github.com/dsfsi/covid19za> (Dataset)

Supplementary material

[10.5334/dsj-2020-019](#) (Journal article)

Data Availability: The data files generated and analyzed in this study are publicly available from the Data Archiving and Networked Services (DANS) database (<https://doi.org/10.17026/dans-zsg-4anh>).

u^b 5. Link your datasets

Reserving a DOI



 **Digital Object Identifier***

Do you already have a DOI for this upload? ☐ Yes ☒ No

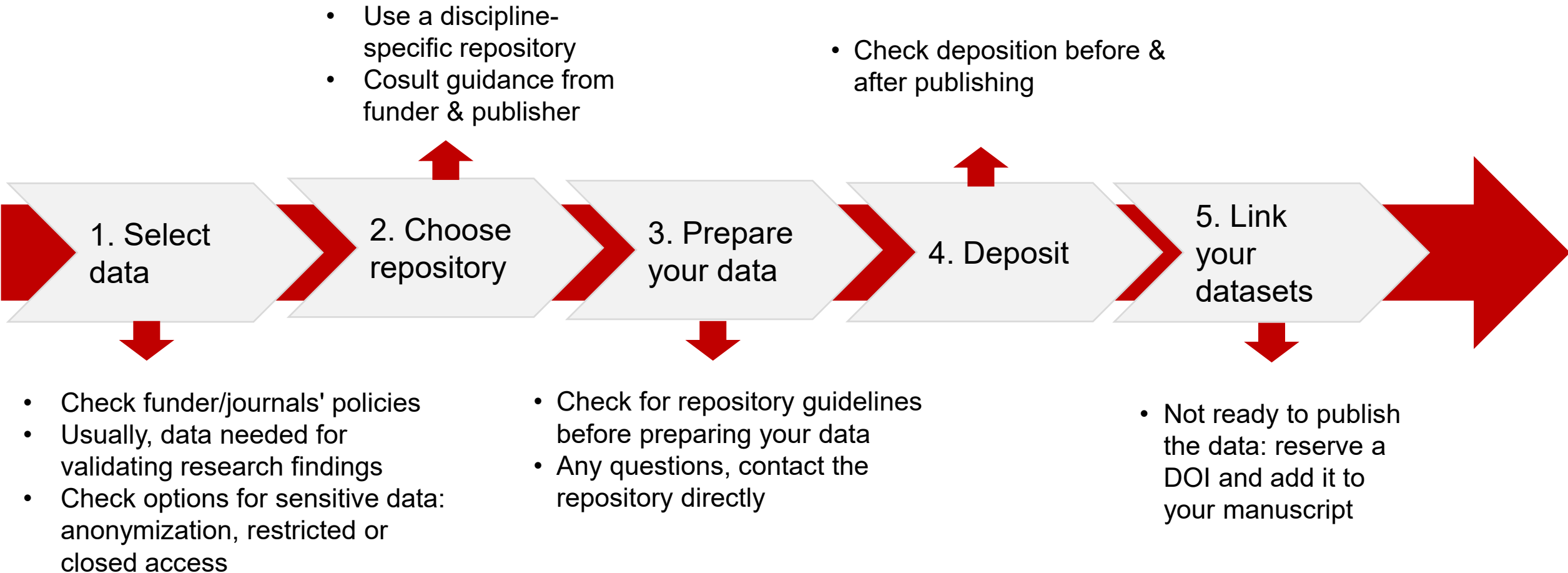
Copy/paste your exi

Get a DOI now!

Reserve a DOI by pressing the button (so it can be included in files prior to upload). The DOI is registered when your upload is published.

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Share Your Data in 5 Steps



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Thank You

For Your Attention

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